

Clinical Text Generation & Interpretability with nanoGPT

Adapting a 10M-Parameter Character-Level Transformer
for Edge-Deployable Healthcare AI Using MIMIC-III

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The Problem: **Healthcare AI is Inaccessible**

- **3.6 billion people** lack access to essential health services (WHO)
- Clinical documentation in LMICs is **paper-based and incomplete**
- Cloud AI (GPT-4, Med-PaLM) requires **internet + expensive subscriptions**
- Patient data sent to **foreign servers** — privacy and sovereignty concerns

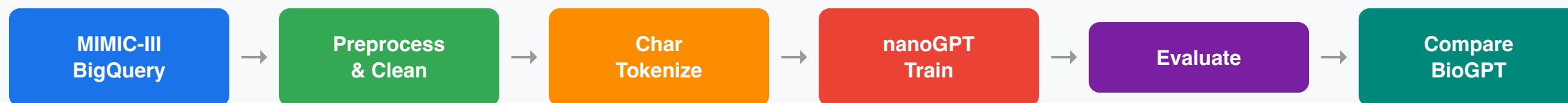
50%+

of the world lacks
essential health services

**Can we build a clinical language model
small enough to run on a \$35 Raspberry Pi,
fine-tune it on local hospital data,
and deploy it as an offline app?**

Our Approach

Built a **10M-parameter GPT from scratch** and trained on **10,000 MIMIC-III discharge summaries**



Model Architecture

Layers / Heads	6 / 6
Embedding dim	384
Context length	256 chars
Parameters	~10M (~40MB)

Key Design Choices

- Character-level tokenization (no BPE)
- Pre-norm Transformer (LayerNorm first)
- Weight tying (embedding = LM head)
- AdamW optimizer, $\text{lr} = 3\text{e-}4$

Training Results

5,000 iterations · 47 minutes · Single NVIDIA T4 GPU

Iteration	Train Loss	Val Loss
0	4.404	4.404
1,000	1.550	1.551
3,000	1.021	1.017
5,000	0.911	0.918

79%

loss reduction
(4.40 → 0.91)

Generated Clinical Text

Prompt: "discharge diagnosis:"

 **Untrained Baseline (random weights)**

```
$%<<y+833*h%u%h']<h%=hu]]^hh2shh>^<%%...
```

 **nanoGPT Clinical (10M, trained)**

```
1. sepsis  
2. sicu disease.  
...
```

 **BioGPT (347M, pretrained)**

```
Discharge diagnosis: schizophrenia or schizoaffective disorder.
```

More Generated Examples

Prompt: "history of present illness: the patient is a"

```
73 year old male with a history of antibiotics and hypertension. she received infection and was started on...
```

Prompt: "medications on admission:"

```
1. colace 100 mg tablet sig: one (1) tablet po q8h (every 6 hours).
```

Prompt: "discharge diagnosis:" (T=0.8)

```
prematurity and coronary artery bypass  
hypertension
```

The model learns medication dosing patterns, diagnosis formatting, and clinical section structure.

Evaluation Metrics

2.52

VAL PERPLEXITY

17

MEDICAL TERMS

0.117

MED TERM FREQ

0.559

TYPE-TOKEN RATIO

0.529

VOCAB COVERAGE

Medical terms found: *discharge, diagnosis, hypertension, tachycardia, heart failure, coronary artery disease, echocardiogram, intracranial hemorrhage, social history, ct head...*

Three-Way Model Comparison

Metric	Untrained Baseline	nanoGPT Clinical (10M)	BioGPT (347M)
Parameters	~10M	~10M	~347M
Tokenization	Char	Char	BPE
Vocab Coverage	0.771	0.543	0.271
Type-Token Ratio	1.000	0.579	1.000
Med Term Frequency	0.065	0.117	0.333
Unique Med Terms	2	11	2

✅ nanoGPT generates **11 unique medical terms** vs. 2 for BioGPT

✅ Med term freq improved **80%** over untrained baseline

⚡ BioGPT (35x larger) achieves 3x higher term frequency

Attention Visualization

Input: "discharge diagnosis: acute myocardial infarction"

Early Layers (Layer 0)

- Strong **local attention** — neighboring characters
- Learning spelling and character patterns
- Diagonal attention pattern

Later Layers (Layer 5)

- **Long-range dependencies** — word and phrase level
- Distributed attention across clinical terms
- Head specialization — different heads learn different patterns

→ The model develops **structured attention patterns** that reveal how it processes clinical language at different levels of abstraction.

The Vision: **Equitable Clinical AI**

- Runs **entirely offline**
- **Fine-tuned locally** by each hospital
- All patient data **stays on-device**
- Auto-completes **discharge summaries**
- Costs **\$50** vs. \$5,000/month

\$50

to deploy per clinic

vs. \$5,000/month for cloud AI

Bringing clinical AI to the **3.6 billion people** who lack access to essential health services.

Future Work

Technical

- **BPE tokenization** for better word-level coherence
- **INT4 quantization** — reduce 40MB → ~10MB
- **Knowledge distillation** from BioGPT
- Scale to full MIMIC-III (~2M notes)

Deployment

- **ONNX/TFLite export** for Raspberry Pi & mobile
- **Mobile app prototype** — offline clinical auto-complete
- **Hospital-specific fine-tuning** — local data, no sharing
- **Differential privacy** — formal guarantees

Thank You

Questions?

MIMIC-III

BigQuery

PyTorch

HuggingFace

Colab Enterprise

T4 GPU

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